

Let's Roll the Dice

Want to have a little fun? How about building a simple dice game? Games are fun, but they are also a great way to learn—especially in Python. Even if you have no intention of building games with your newfound programming knowledge, the skills you learn while building the next few games can easily translate to other types of apps you might want to build.

The dice game you will develop uses a function called `randint()` that is not included in Python. That means it needs to be imported. You can import entire modules, but they are very big. They contain many functions—the one you need, as well as many you probably do not need. In the interest of keeping your apps running quickly and smoothly, it is best to import only the functions you need, rather than the entire module. To import the `randint()` function from the `random` module, type this:

```
from random import randint
```

While you are at it, import one more function—the `sleep()` function from the `time` module. The `sleep()` function pauses script execution for a designated amount of time, measured in seconds, so that the processes in the script do not all execute instantly. You want to simulate the rolling of dice, after all, and that takes a few seconds.

```
from time import sleep
```

Now you can simulate the rolling of a pair of dice. To keep this a little more interesting, have the computer roll the dice three

times and print out all three rolls. First, let the user know what the game is all about:

```
print("It is time to roll the dice three times and see which  
roll is the highest! Are you ready?")
```

Then, define a function called `roll_dice()` using `def`. The `roll_dice()` function takes an argument of the total number possible from a roll, which is 12 (given that each die has a maximum value of 6 and there are two dice). The code below is commented for explanatory purposes:

```
def roll_dice(number_of_sides):  
#For each roll, randint takes arguments of a minimum  
value of 2 and a maximum value of number_of_sides be-  
cause no roll can be less than 1  
    first_roll = randint(2,number_of_sides)  
    second_roll = randint(2,number_of_sides)  
    third_roll = randint(2,number_of_sides)  
#The sleep function takes a number of seconds as its ar-  
gument  
    sleep(2)  
#After two seconds of sleep, print the result of the roll  
    print("The first roll is " + str(first_roll))  
    sleep(2)  
    print("The second roll is " + str(second_roll))  
    sleep(2)  
    print("The third roll is " + str(third_roll))
```

You defined the function above and now you can call the function to launch the program. Be sure to include the maximum value of 12 as an argument here:

roll_dice(12)

After the function has finished, print a thank you message to the player:

print("Thanks for playing!")

There is a lot going on here, but it should all look fairly familiar. This is not exactly the most exciting game ever written, but using your imagination, you can come up with a lot of fun ways to use the randint function for games! Just make sure you import it in each program.